

Generative AI for Discovery (AI4Science)

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My Research Statement

Help people find, verify
and organize Information
for Complex Information
Needs

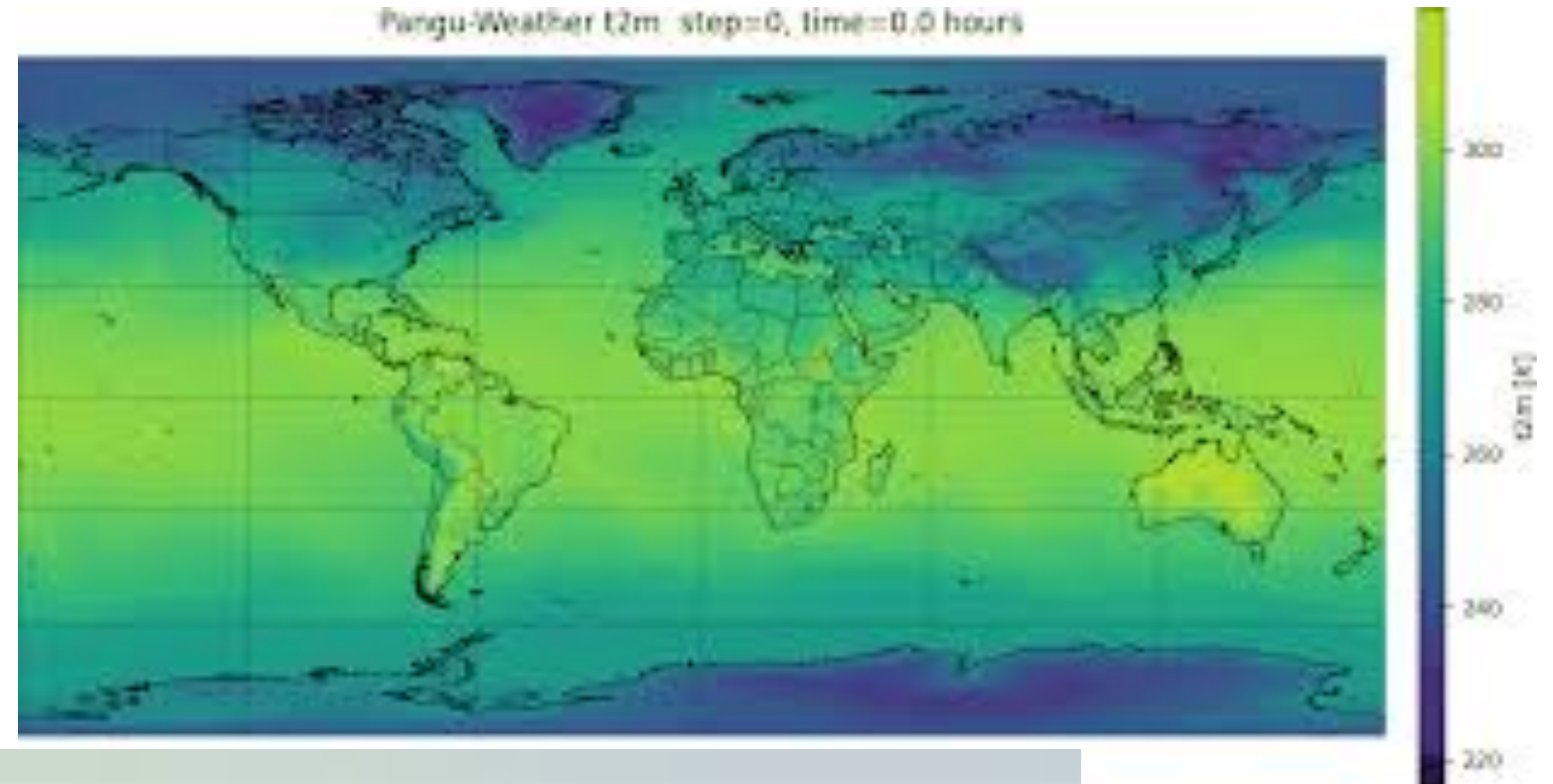
“with modest resources”

Agenda

- AI4Science
- Alphafold
- Discussion of Alphafold 3
- AutoDiscovery - How it works?
- GenAi4MaterialScience

Why GenAI4Science?

- Is not just limited to prompting LLMs
- Has huge potential to solve open problems in climate, healthcare, physics by modeling real-world dynamics
- Has the potential to win you the Nobel prize.



AlphaFold 2

 DeepMind
AlphaFold AI



<https://www.youtube.com/watch?v=d95J8yzvjbQ>

AlphaFold3

AlphaFold 3 (AF3) represents a significant evolution from AlphaFold 2 (AF2), shifting from a protein-centric model to a general biomolecular modeling system. While AF2 focuses on protein structures, AF3 offers broader capabilities, including the modeling of DNA, RNA, ligands, and post-translational modifications with higher accuracy

AF3 is trained on molecular structural data contained within the protein databank and can process 99% of all known bimolecular complexes

Broad architecture of AF3

- AlphaFold3 (AF3) is a generative AI model composed of a custom Transformer model (a deep neural network) with triangular attention mechanism. It is an improvement of previous model AlphaFold2.
- It takes as input a description of a biomolecular complex (a collection of multiple proteins, DNA, RNA, molecular ligands, ions) and generates/ predicts the 3D structure of that biomolecular complex.

It uses **diffusion process** to generate individual 3D coordinates of every atom in the input-specified system

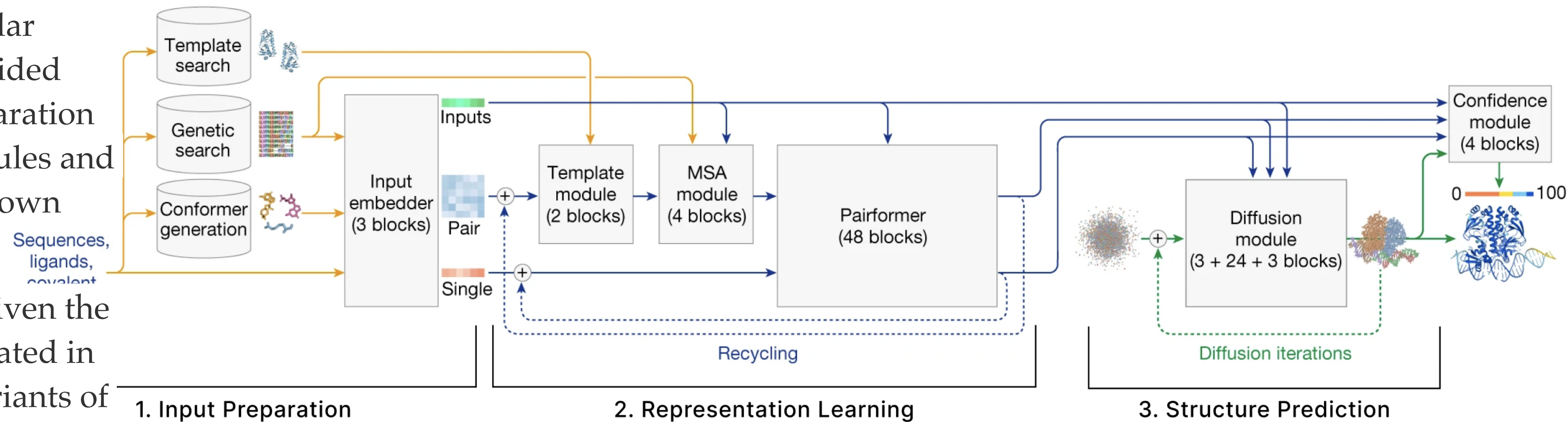
Broad architecture of AF3

1. Input Preparation The user provides sequences of some molecules to predict structures for and these need to be embedded into numerical tensors.

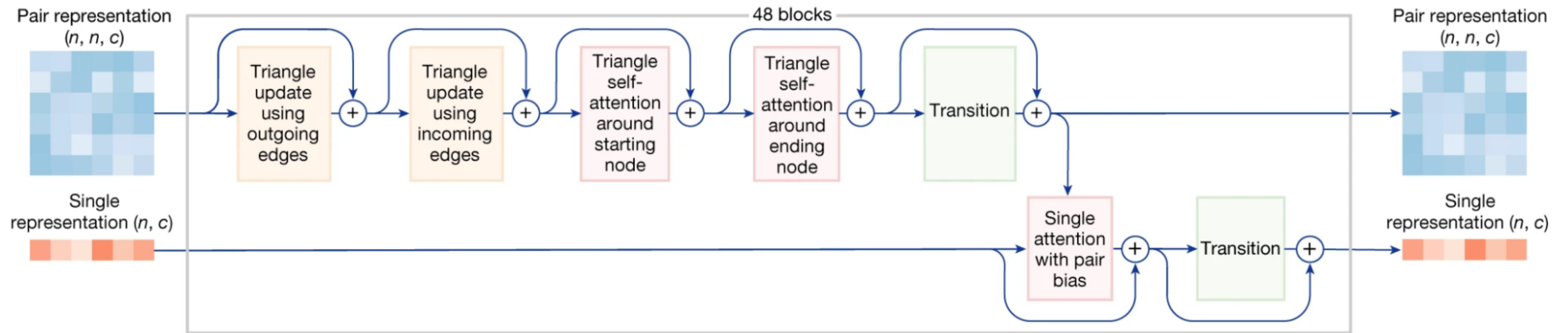
Furthermore, the model retrieves a collection of other molecules that are presumed to have similar structures to the user-provided molecules. The input preparation step identifies these molecules and also embeds these as their own tensors.

2. Representation learning Given the Single and Pair tensors created in section 1, we use many variants of attention to update these representations.

3. Structure prediction We use these improved representations, and the original inputs created in section 1 to predict the structure using conditional diffusion.



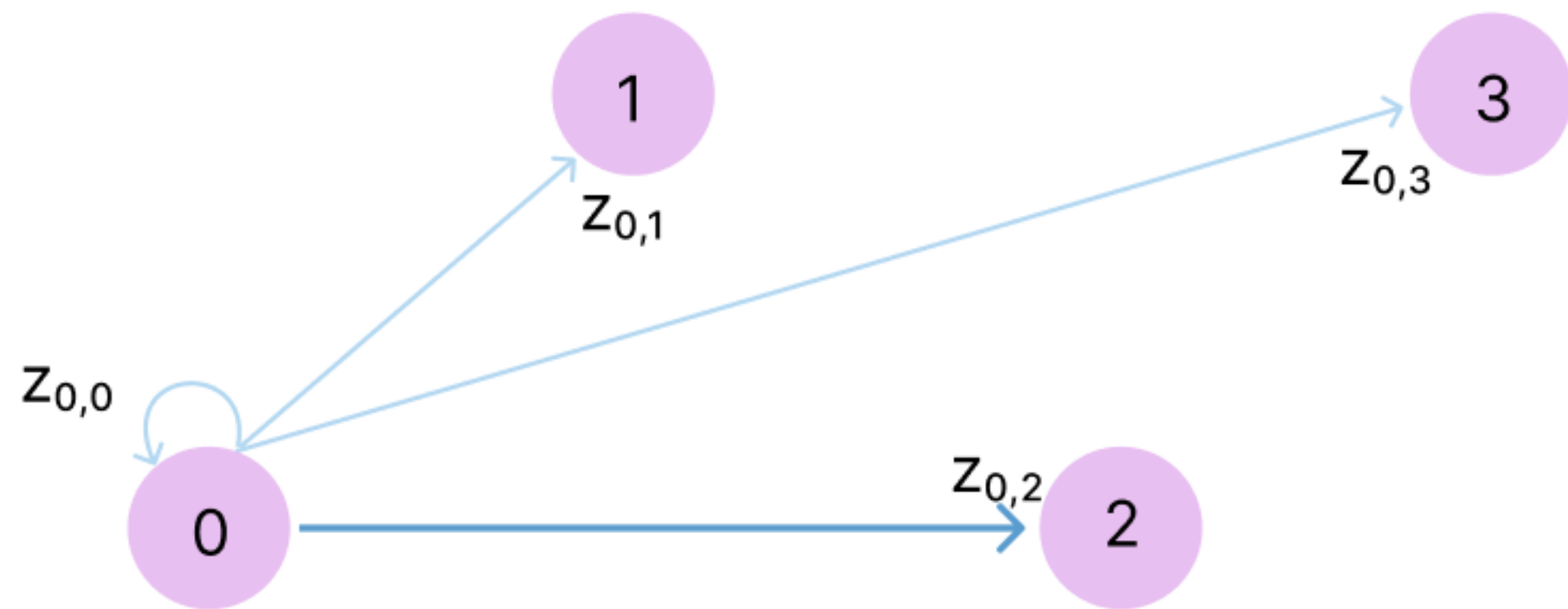
Pairformer in Alphafold3



Performer updates pairwise and single representation using triangle attention

It is also a representation learning module

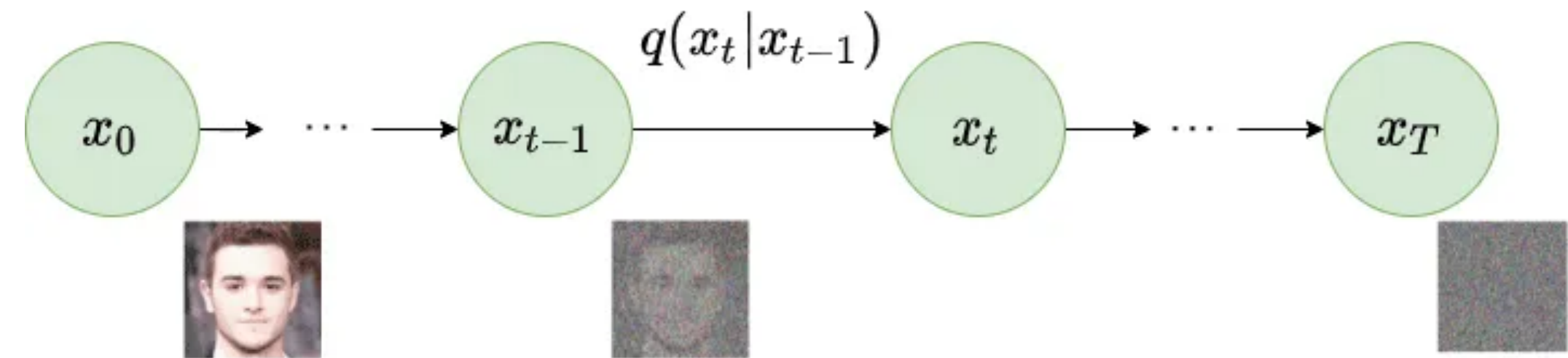
Why Triangles?



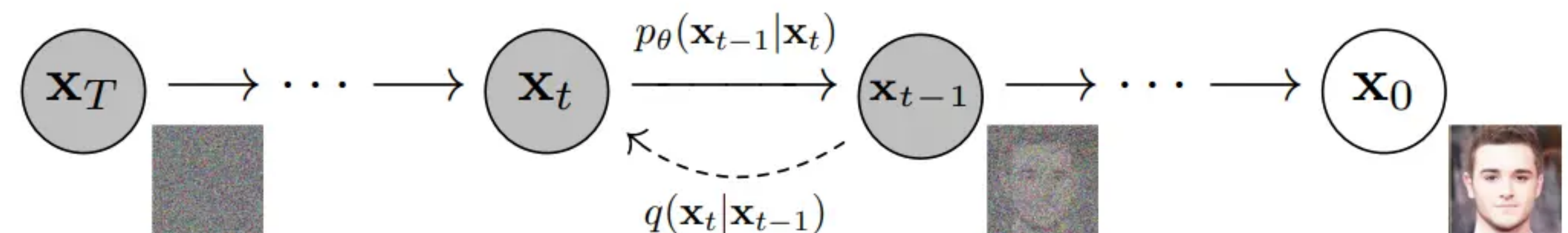
The guiding principle here is the idea of the triangle inequality: “the sum of any two sides of a triangle is greater than or equal to the third side”

Recap Diffusion?

Noise is iteratively added to the data over a series of T timesteps to create a sequence of T variants of each datapoint. We call the original data point $x_{t=0}$ and the fully noised version $x_{t=T}$. During training, at timestep t , the model is given the x_t and predicts what noise was added between x_{t-1} and x_t . We take a gradient step on the predicted noise added compared to the actual noise that had been added.



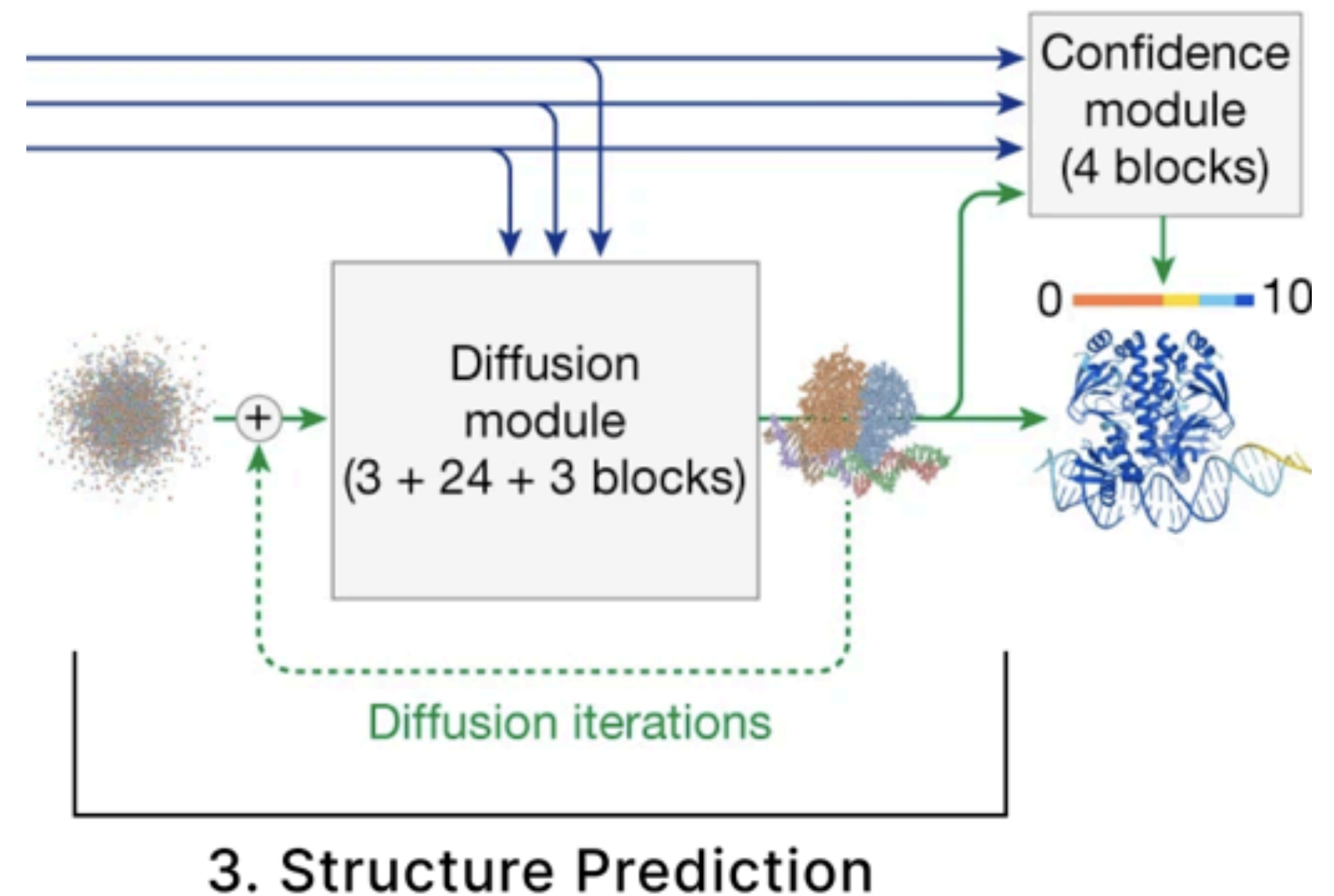
Then, at inference time, we simply start with random noise, which is equivalent to $x_{t=T}$. For every time step, we predict the noise the model thinks has been added, and remove that predicted noise. After a pre-specified number of timesteps, we end up with a fully “denoised” datapoint that should resemble the original data from our dataset.



Conditional Diffusion in Alphafold3

Conditional Diffusion lets the model ‘condition’ these de-noising predictions on some input. Practically this means that for each step of the model, it takes three inputs:

1. The current noisy iteration of our generation
2. A representation of the current time step we are at
3. The information we want to condition on (this could be a caption for an image to generate, or properties for a protein)

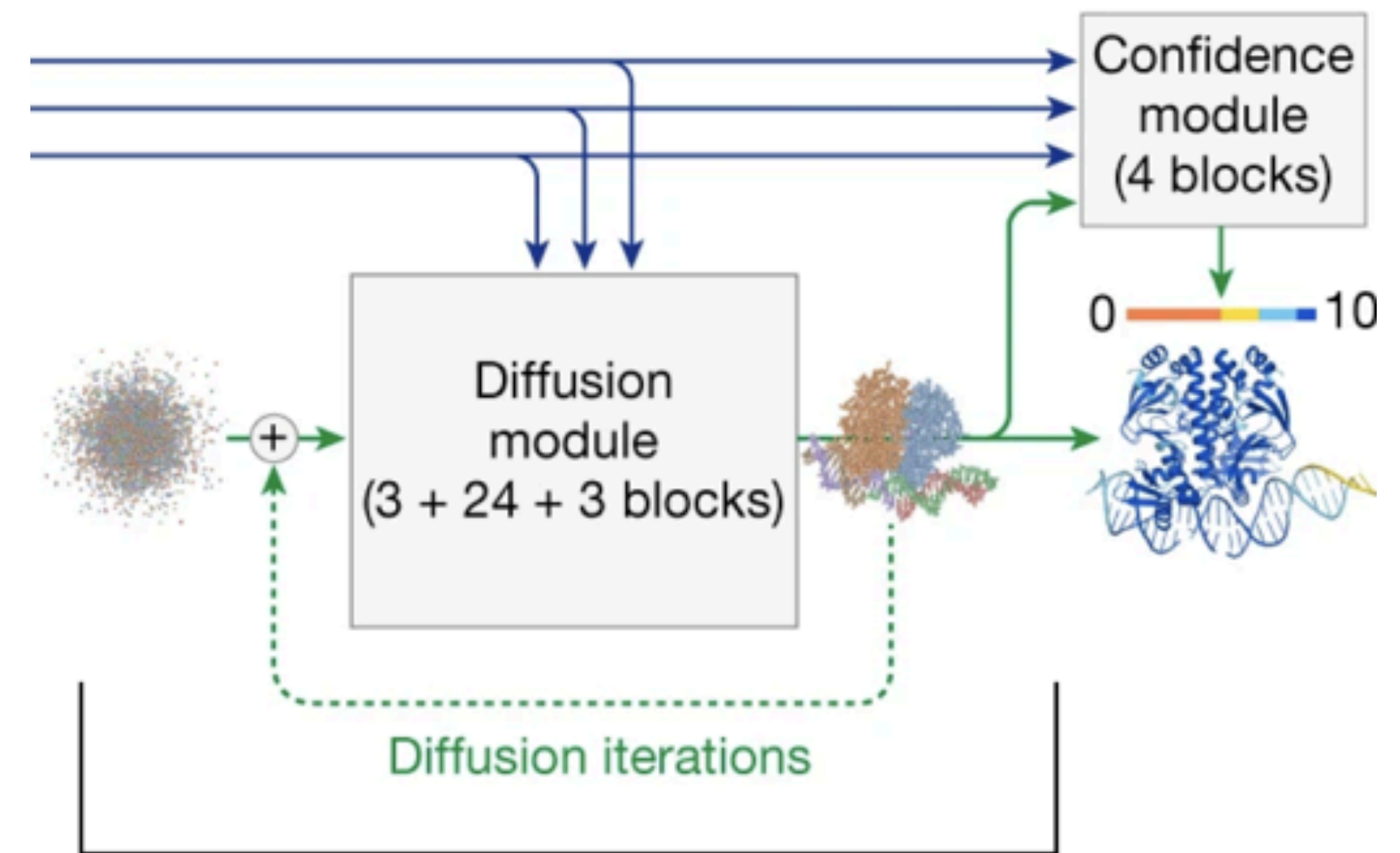


The final generation is not just a random example that resembles the training data distribution, but should specifically match the information represented by this conditioning vector.

Conditional Diffusion in Alphafold3

- With AF3, the data we learn to de-noise is a matrix $\mathbf{x}$ with the x,y,z coordinates of all the atoms in our sequences.
- During training, we add Gaussian noise to these coordinates until they are effectively fully random. Then at inference time, we start with random coordinates.
- At each time step, AF3 first randomly rotate and translate our entire predicted complex. This data-augmentation teaches the model that any rotation and translation of the complex is equally valid.

Af3 then adds a small amount of noise to the coordinates to encourage more heterogeneous generations. Finally, we predict a de-noising step using the Diffusion Module



3. Structure Prediction

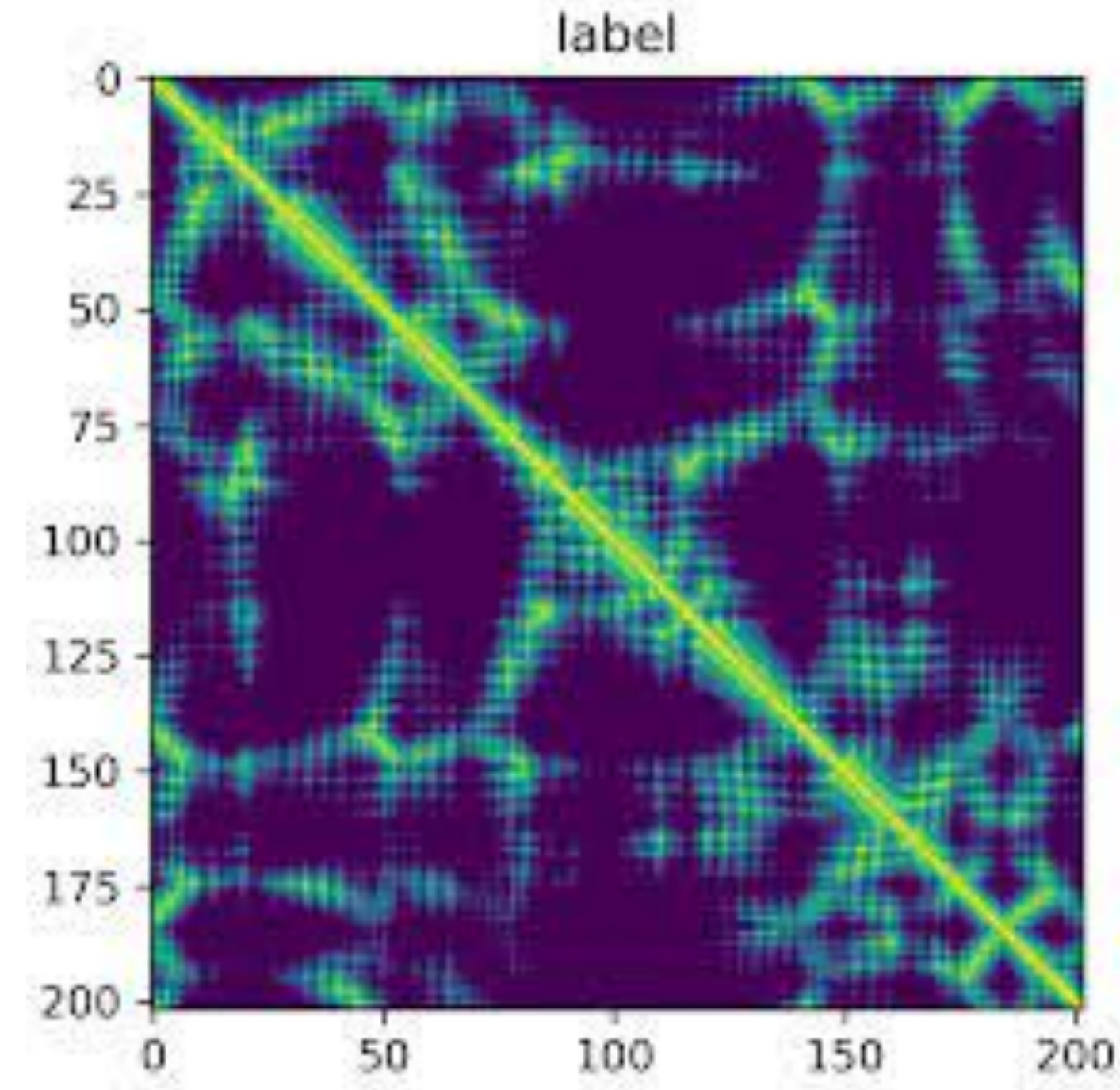
Loss function

$$L_{\text{loss}} = L_{\text{distogram}} * a_{\text{distogram}} + L_{\text{diffusion}} * a_{\text{diffusion}} + L_{\text{confidence}} * a_{\text{confidence}}$$

- **Ldistogram** which evaluates the accuracy of the predicted distogram at a token-level
- **Ldiffusion** which evaluates the accuracy of the predicted distogram at an atom-level. It looks at all pairwise distances then includes additional terms to prioritize distances between nearby atoms and atoms involved in protein-ligand bonds.
- **Lconfidence** which evaluates the model's self-awareness about which structures are likely to be inaccurate

Distogram Loss

- The output of the model is atom-level coordinates, and this loss evaluates a token-level distogram. It is a 2D matrix that predicts the probability distribution of distances between all pairs of amino acid residues
- To get the xyz coordinates for tokens, we just use the coordinate of the “center atom”.
- As these distogram distances are categorical, the predicted distogram is then compared to the true distogram via cross entropy



Diffusion Loss

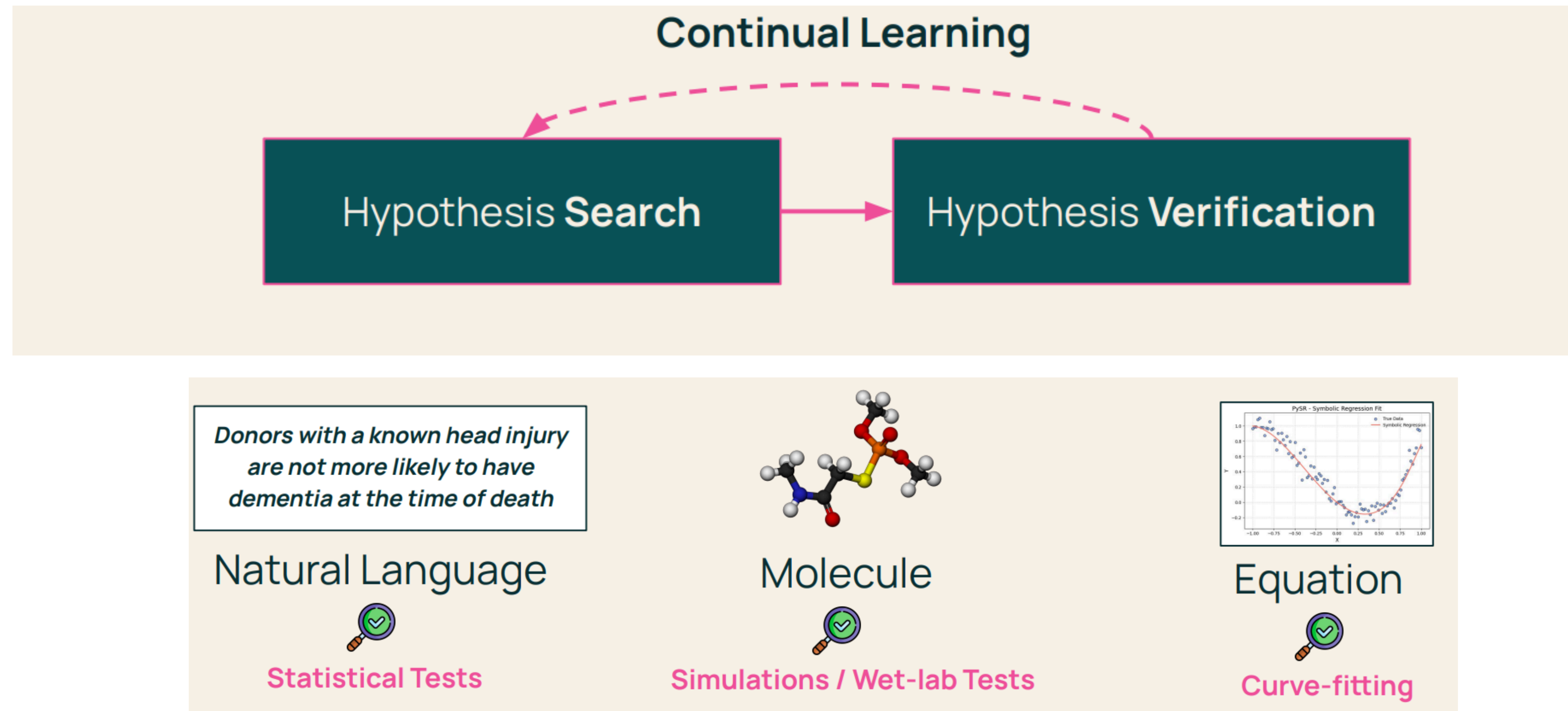
$$L_{\text{diffusion}} = (L_{\text{MSE}} + L_{\text{bond}} * \alpha_{\text{bond}}) * (\hat{r}^2 + \sigma_{\text{data}}^2) / (\hat{r} + \sigma_{\text{data}})^2 + L_{\text{smooth_lddt}}$$

- **LMSE** is a version of the distogram loss, but over all atoms rather than just “center atoms”. Additionally, it looks at the mean squared error between positions, rather than binning them into a distogram.
- **Lbond** aims to ensure the accuracy of bond lengths for protein-ligand bonds by adding an additional MSE loss on the difference in predicted and ground-truth distograms for atom-pairs that are part of protein-ligand bonds.
- **Lsmooth_LDDT** (smoothed local distance difference test) is yet another variant of the distogram loss that tries to capture the accuracy of local distances. An atom-pair’s predicted distance “passes the test” if it is within a given threshold of the atom-pair’s true distance. Pass through sigmoid to make it smooth

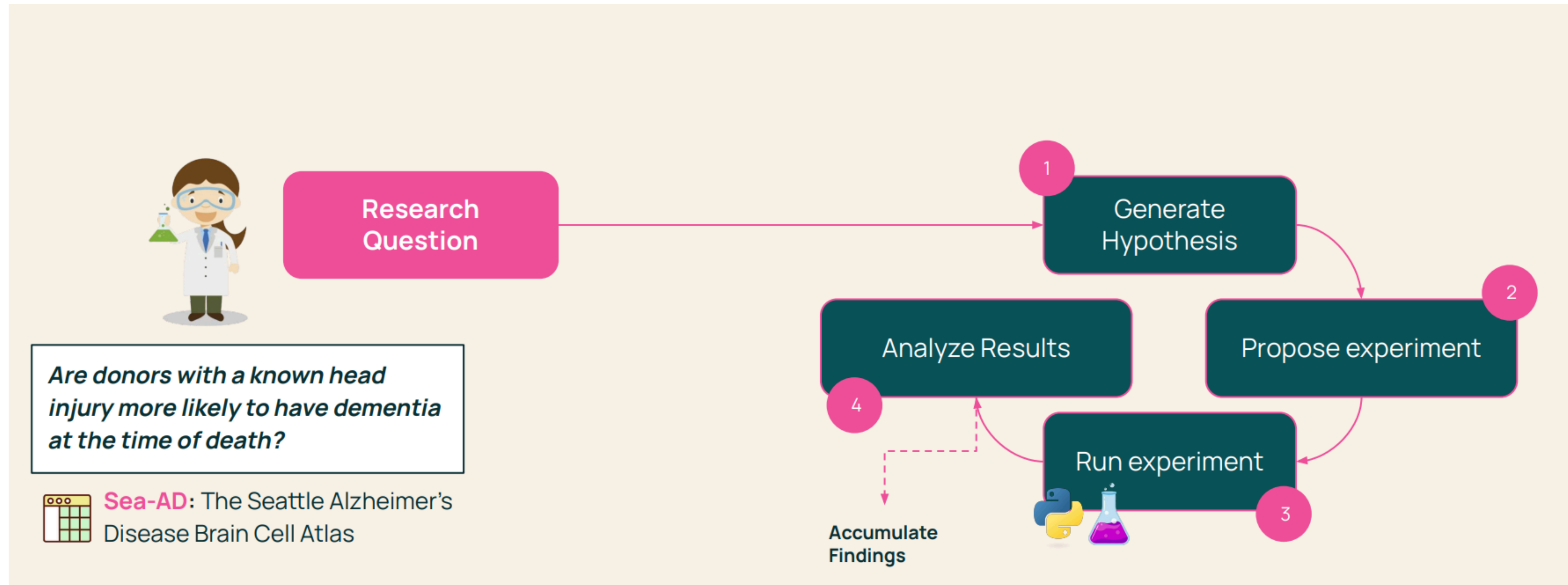
What do you think is the recipe for discovery ?

1. Study of existing methods and come up with a combination of them and derive insights.
2. Formulate a hypothesis based on knowledge of the field, search more literature, borrow ideas / try ideas exhaustively, identify gaps through experimentation repeat.
3. Formulate a hypothesis based on knowledge of the field, search more literature, borrow ideas / try ideas, focus on ideas with surprising results through experimentation repeat.

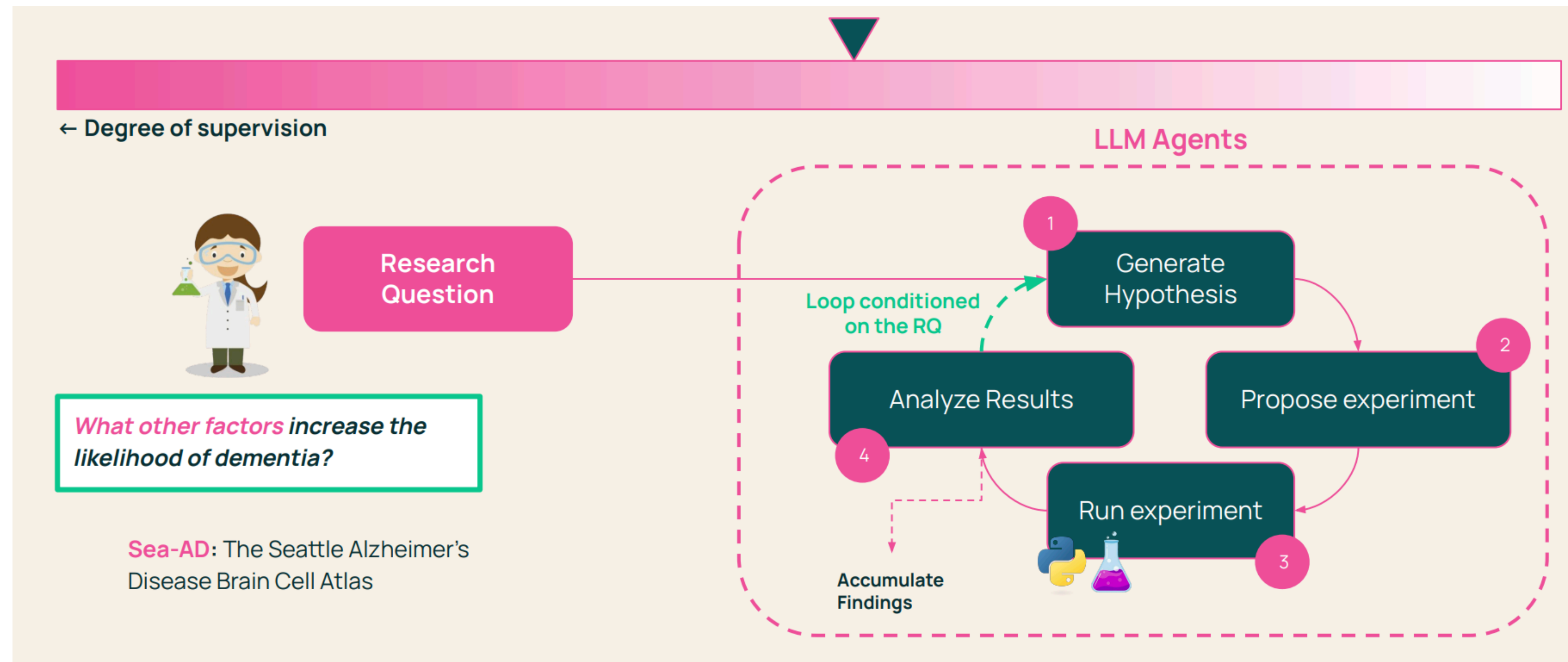
What is the recipe for discovery ?



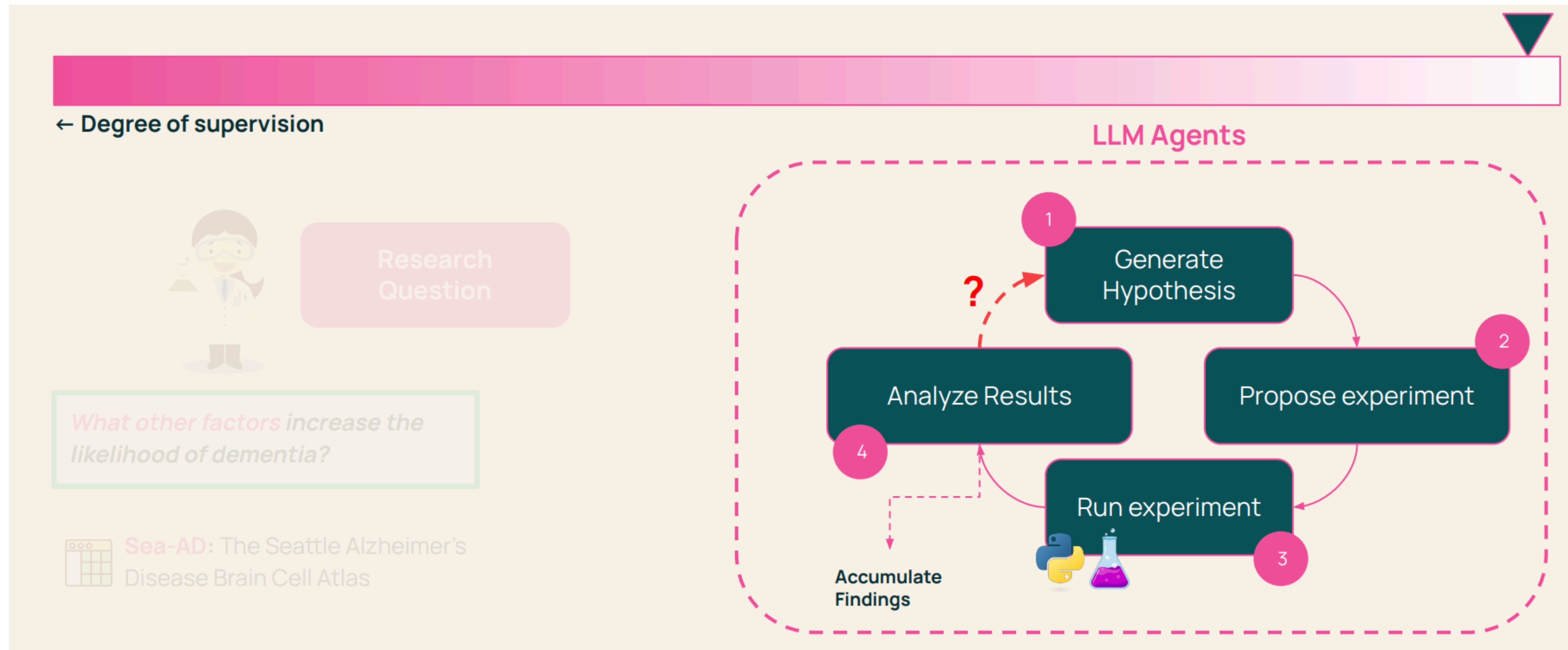
How is it done today?



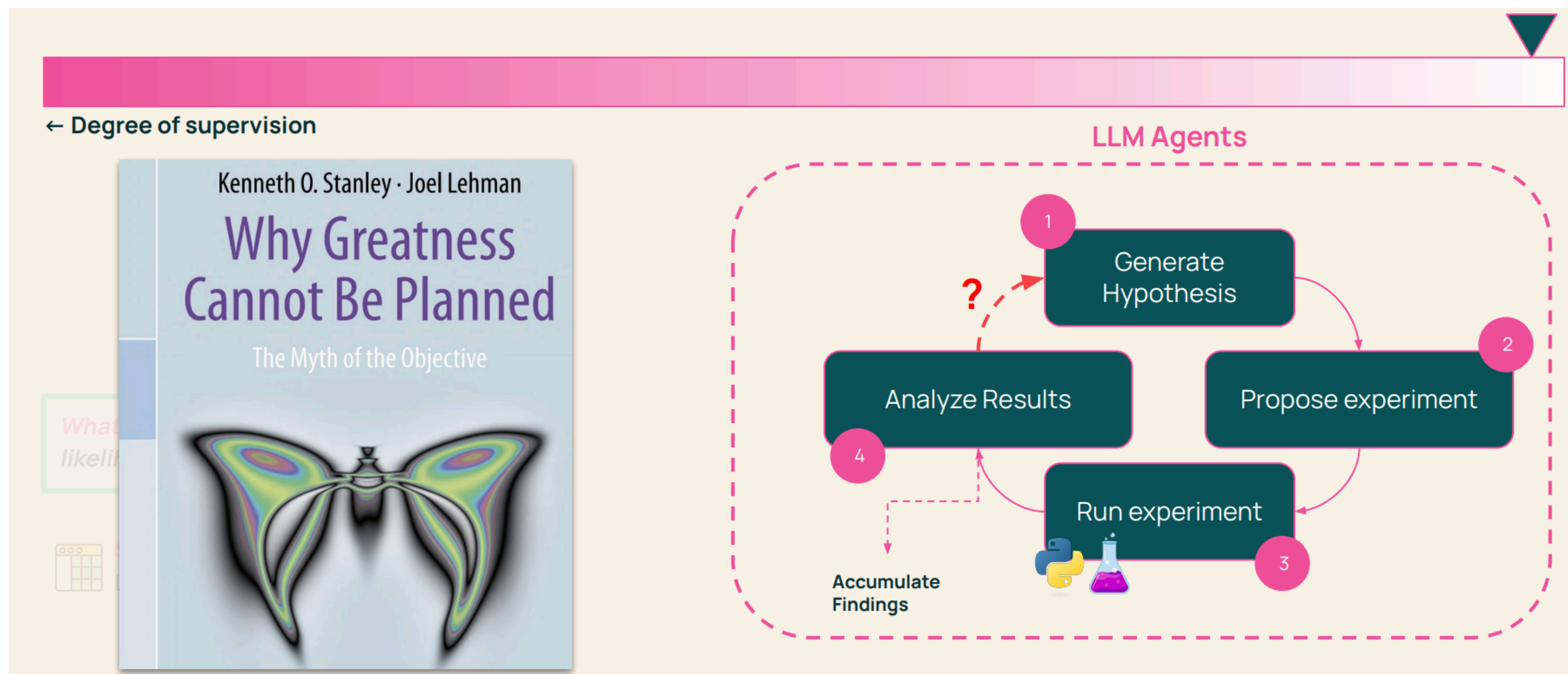
Can LLM agents help?



What if no-one supplies the central RQ?



We need to simulate open-ended exploration



Enter bayesian surprise

$$D_{KL}(\text{posterior} || \text{prior})$$

After experimental evidence

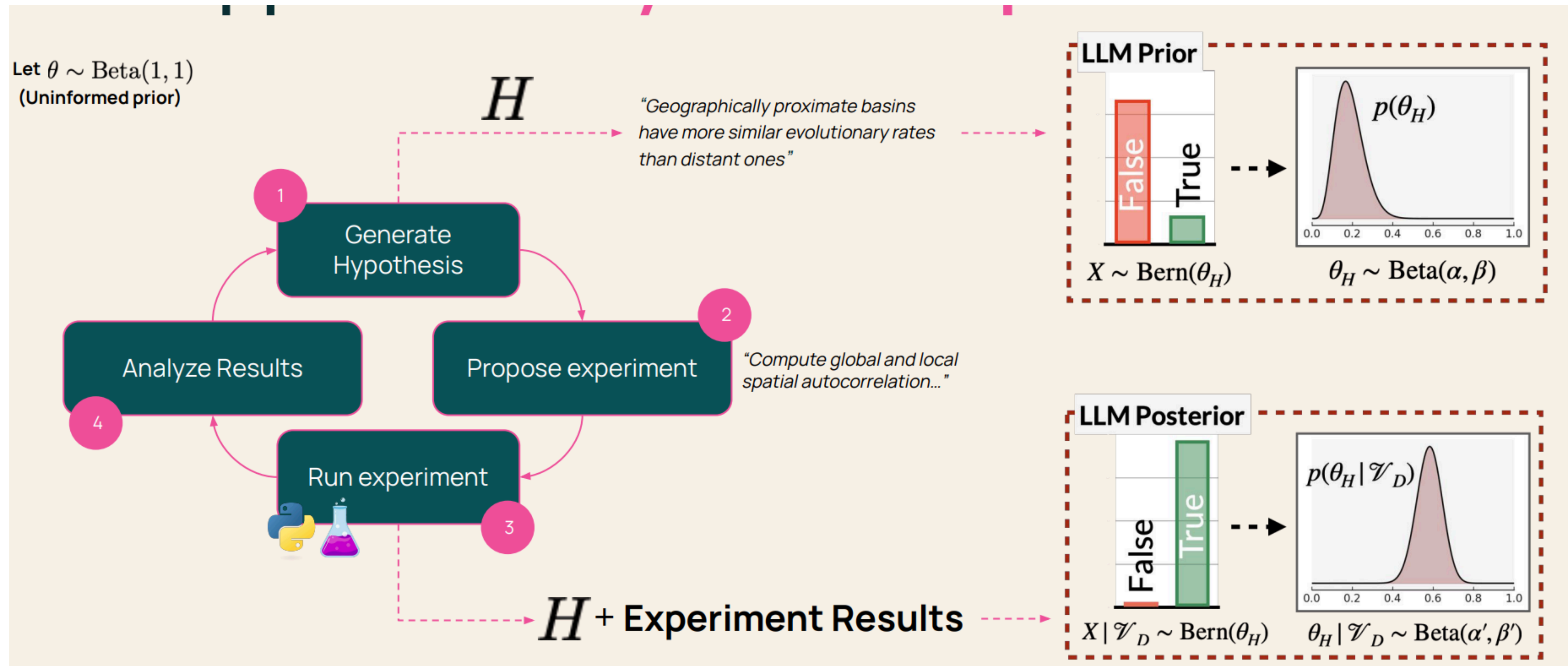
Before experimental evidence

Agent's belief about a hypothesis H:

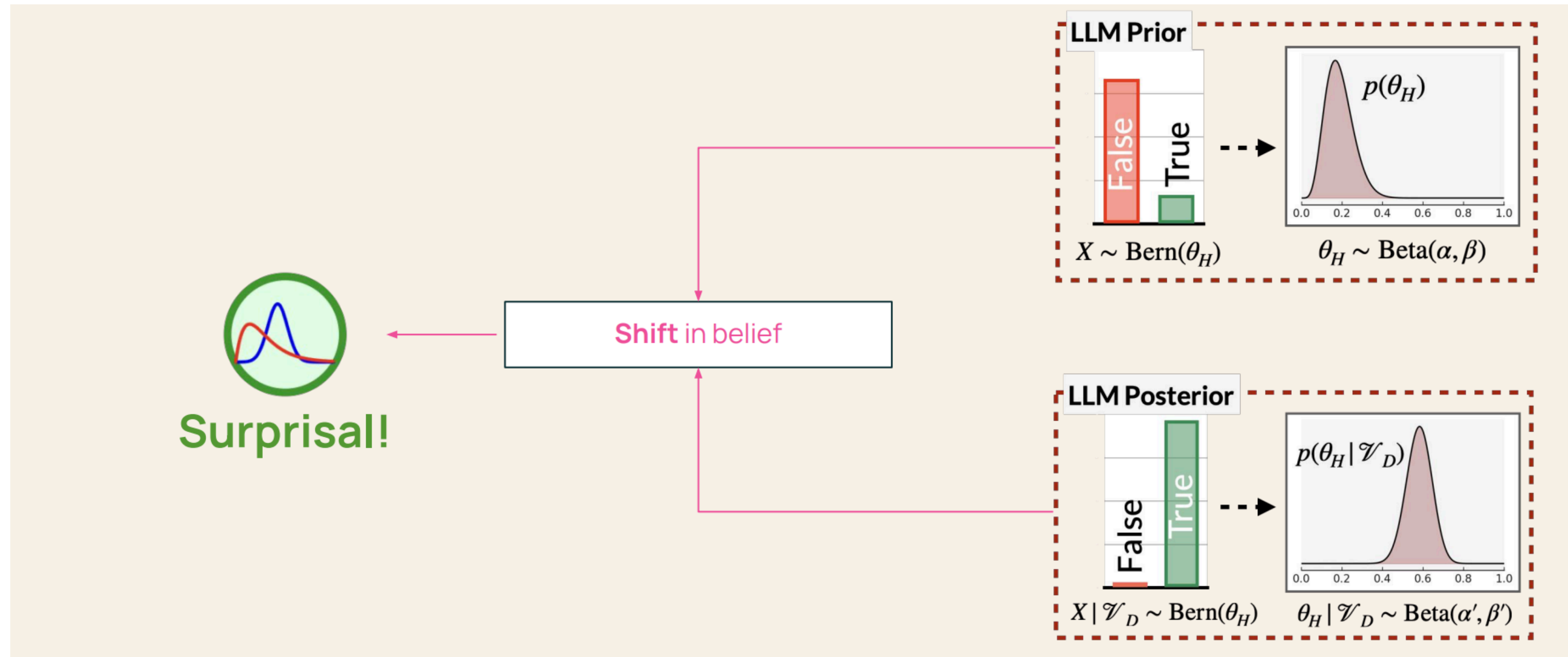
Distribution (capturing uncertainty) over probabilities that a given hypothesis is true.

$$\theta_H \in [0, 1]; \theta_H \sim \text{Beta}(\alpha, \beta)$$

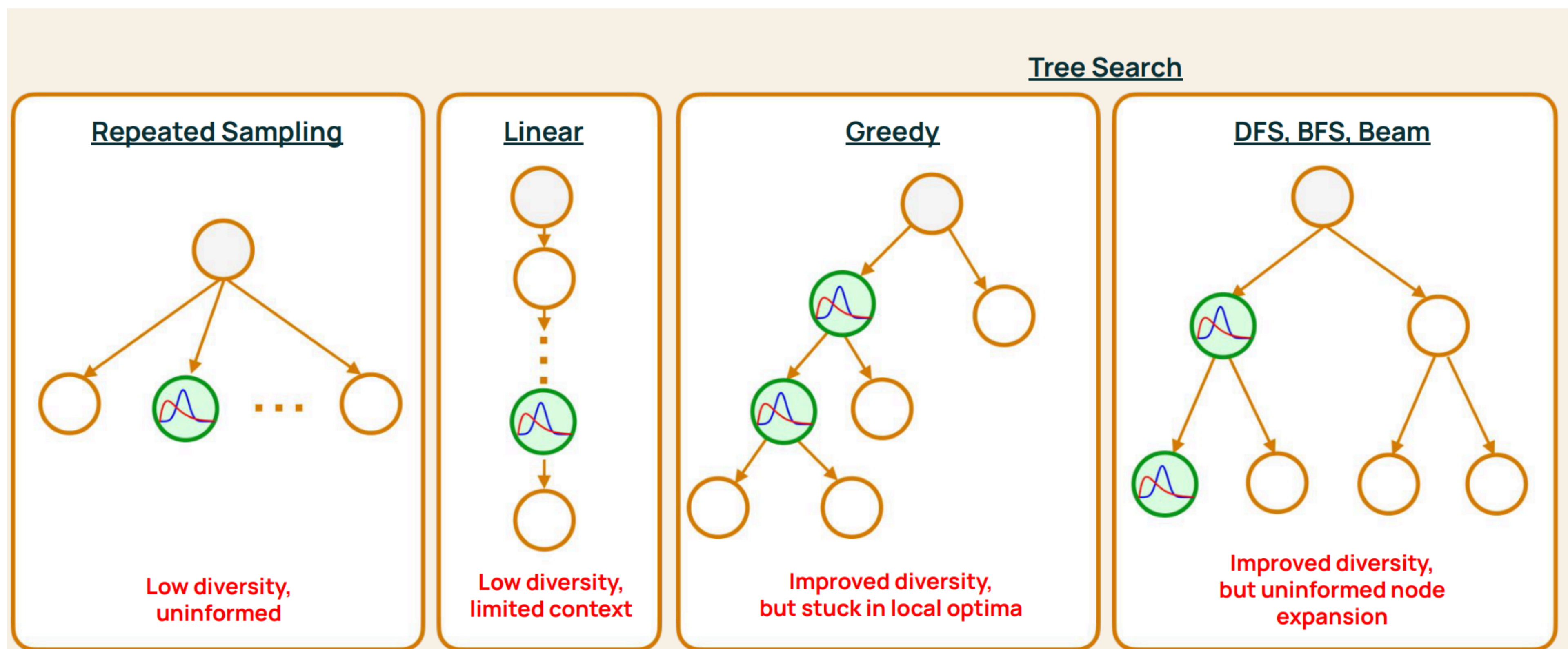
Enter bayesian surprise



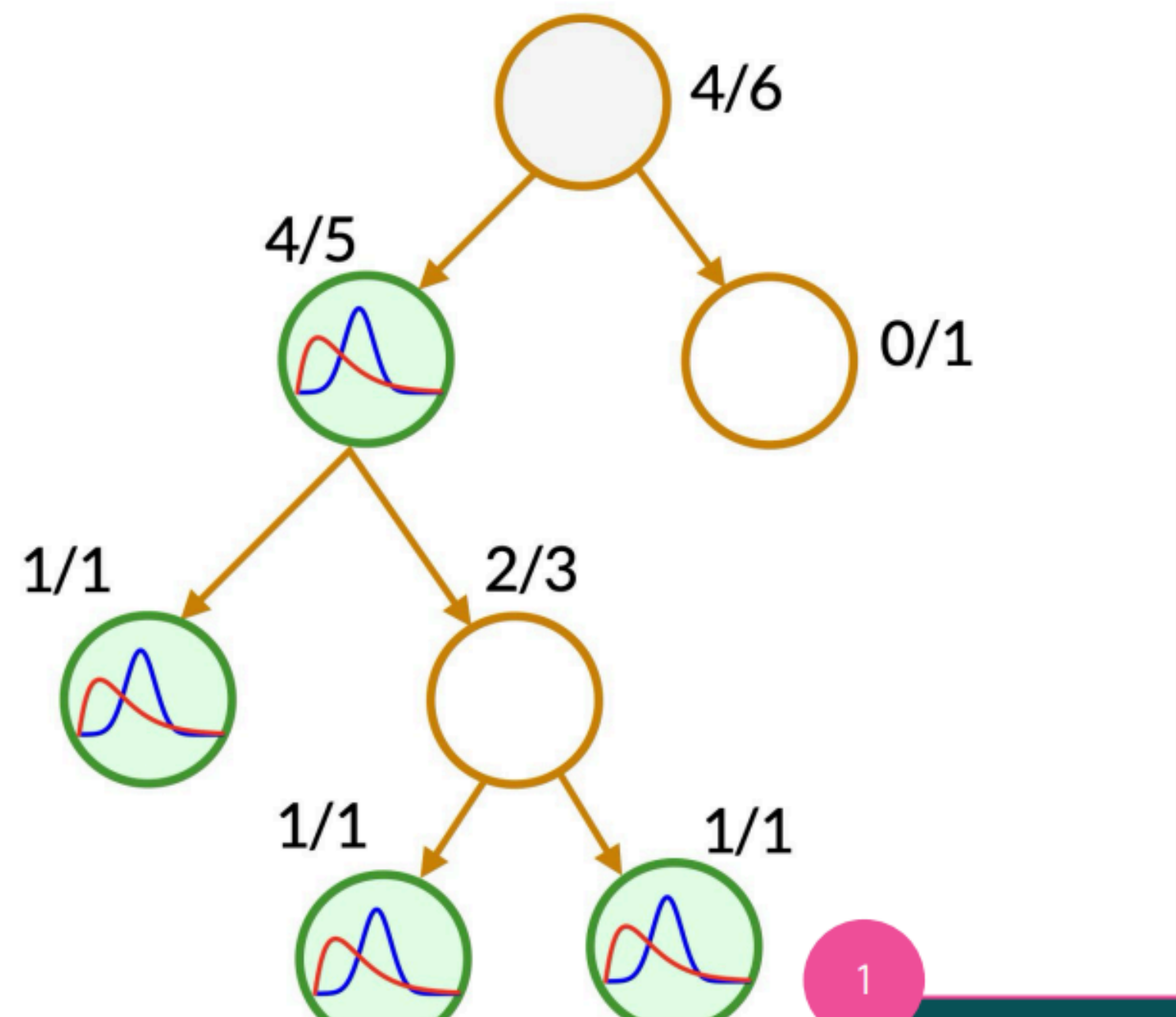
Enter bayesian surprise



Search in solution space for discovery



MCTS (Monte-carlo tree search)



Reward: Bayesian surprise.

Nodes in the search tree with high surprisal counts are likely to elicit surprisal on further expansion

Caveat:

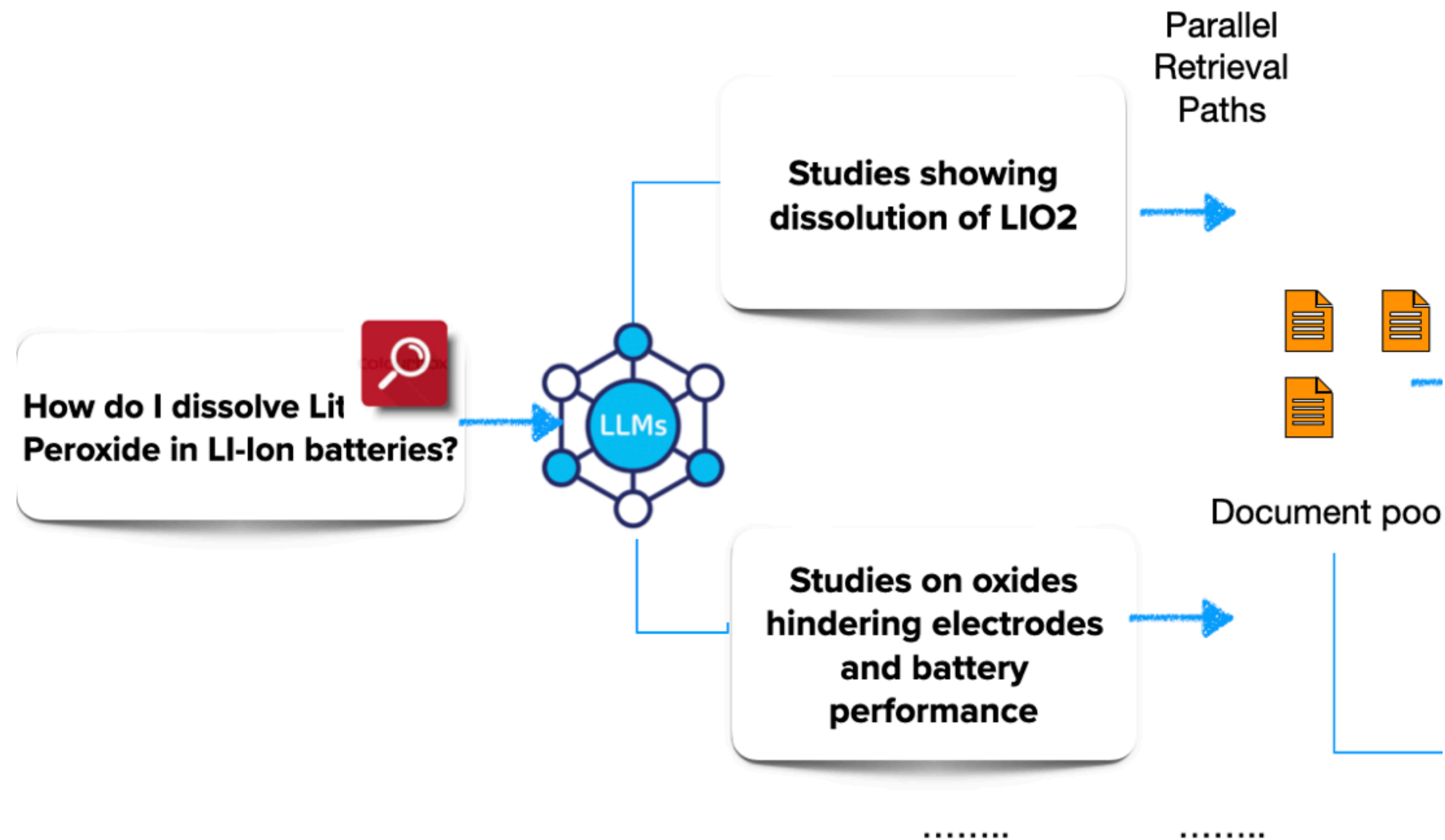
The rollouts of nodes are expensive!!

Helping Material Scientists

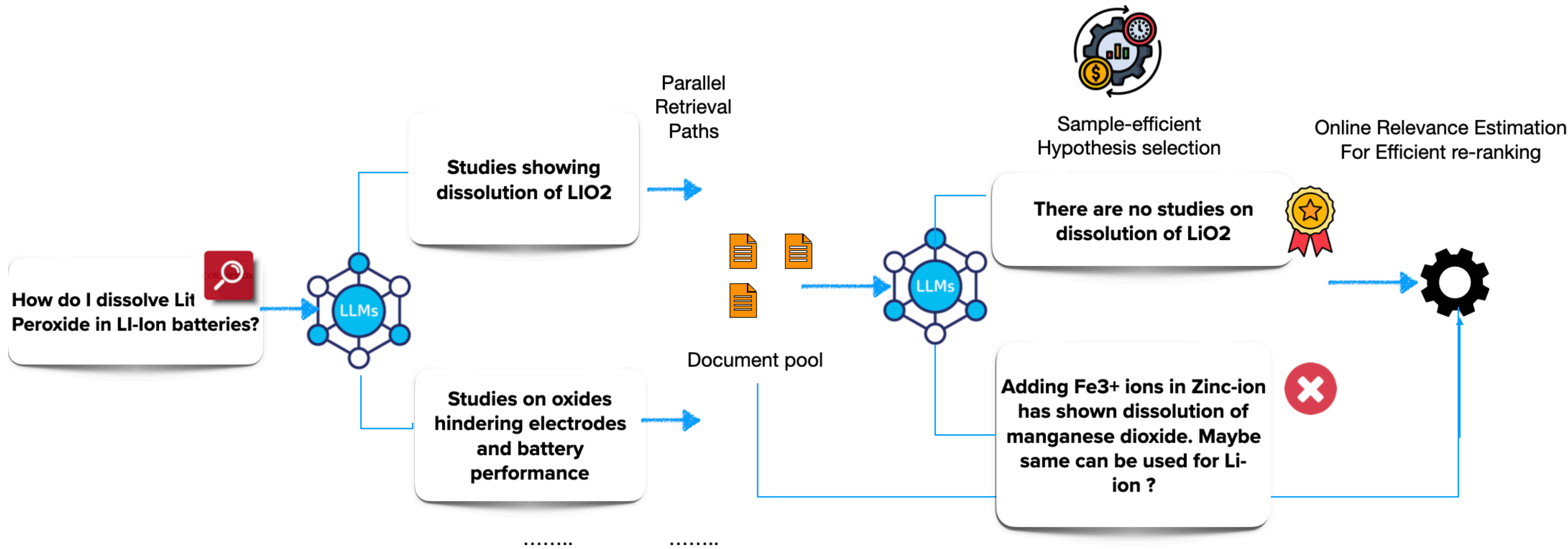
**How do I dissolve Lit
Peroxide in LI-Ion batteries?**



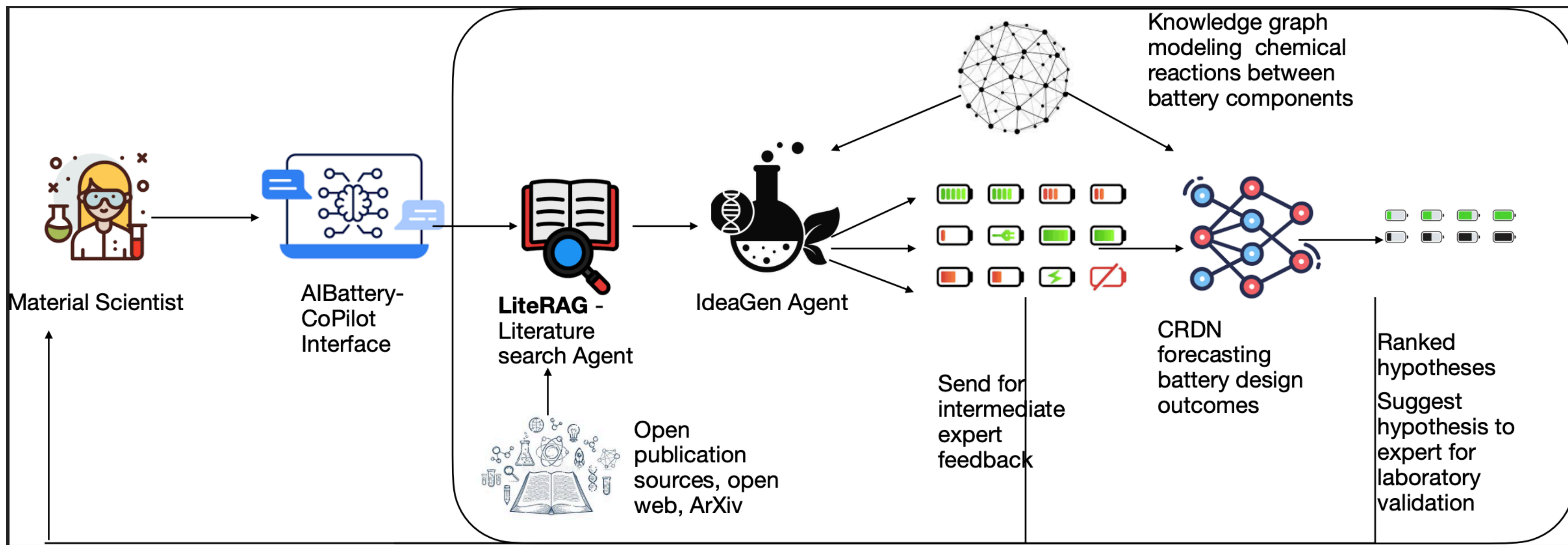
Helping Material Scientists - Query generation & Retrieval



Helping Material Scientists



AI-battery-Copilot



RESOURCES

<https://www.youtube.com/watch?v=d95J8yzvjbQ>

<https://allenai.org/blog/autodiscovery>

<https://www.majumderb.com/AutoDiscovery.pdf>

Reference: <https://elanapearl.github.io/blog/2024/the-illustrated-alphafold/>

Thank you !

Questions ?